Scalar, Vector, Matrix : Algebra

Scalar: $a = 3 = (3)$

Vector: $\vec{a} = (2,\ 4),\ \ \vec{a} = (2,\ 3,\ 1), \cdots$

$$\vec{b} = \begin{pmatrix} 1 \\ 4 \end{pmatrix},\ \ \vec{b} = \begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix}, \cdots$$

Notation rule of vector: $\mathbf{a},\ \boldsymbol{a},\ \vec{a}$

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Row vector

$$\mathbf{a} = (2,\ 4),\ \ \mathbf{a} = (2,\ 3,\ 1)$$

Transposed vector

$$\mathbf{a}^{\mathrm{T}} = \begin{pmatrix} 2 \\ 4 \end{pmatrix},\ \ \mathbf{a}^{\mathrm{T}} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

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Column vector

$$\mathbf{b} = \begin{pmatrix} 5 \\ 2 \end{pmatrix},\ \ \mathbf{b} = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$$

Transposed vector

$$\mathbf{b}^{\mathrm{T}} = (5,\ 2),\ \ \mathbf{b}^{\mathrm{T}} = (5,\ 2,\ 1)$$

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$$\mathbf{a} = (2,\ 4), \mathbf{b} = (5,\ 2)$$

$$\mathbf{a} + \mathbf{b} = (2,\ 4) + (5,\ 2) = (7,\ 6)$$

$$\mathbf{a} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$$

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$$a = 3, \mathbf{b} = (3,\ 2)$$

$$a\mathbf{b} = 3 \times (3,\ 2) = (9,\ 6)$$

$$a = 3, \mathbf{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$a\mathbf{b} = 3 \times \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 9 \\ 6 \end{pmatrix}$$

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Matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}$$

$$\mathbf{A} = \left(\begin{array}{cc} 1 & 3 \\ \hline 2 & -1 \end{array}\right)$$

$$\mathbf{A} = \left(\begin{array}{c|c} 1 & 3 \\ 2 & -1 \end{array}\right)$$

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Matrix element (Scalars)

$$\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}$$

The element in the first row and second column is written as a_{12}.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$a_{12} = 3.$

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Matrix element (Vectors)

Matrix $\mathbf{A}$ consists of $\mathbf{a}_1 = (a_{11} \;\; a_{12})$, $\mathbf{a}_2 = (a_{21} \;\; a_{22})$, and expressed as

$$\mathbf{A} = \begin{pmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \end{pmatrix} = \begin{pmatrix} (a_{11} & a_{12}) \\ (a_{21} & a_{22}) \end{pmatrix}$$

Matrix $\mathbf{B}$ consists of $\mathbf{b}_1 = \begin{pmatrix} b_{11} \\ b_{21} \end{pmatrix}$, $\mathbf{b}_2 = \begin{pmatrix} b_{12} \\ b_{22} \end{pmatrix}$ and expressed as

$$\mathbf{B} = (\mathbf{b}_1 \;\; \mathbf{b}_2) = \left(\begin{pmatrix} b_{11} \\ b_{21} \end{pmatrix}, \begin{pmatrix} b_{12} \\ b_{22} \end{pmatrix} \right)$$

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Matrix element (Matrixes)

$$\mathbf{A} = \left(\begin{array}{cc|ccc} 1 & 4 & 0 & 0 & 2 \\ 1 & 4 & 0 & -3 & 2 \\ 2 & 3 & 0 & 0 & 0 \\ \hline 4 & 0 & 1 & 2 & 0 \\ 0 & 0 & 7 & 1 & 0 \end{array} \right)$$

$$= \begin{pmatrix} \begin{pmatrix} 1 & 4 \\ 1 & 4 \\ 2 & 3 \end{pmatrix} & \begin{pmatrix} 0 & 0 & 2 \\ 0 & -3 & 2 \\ 0 & 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} & \begin{pmatrix} 1 & 2 & 0 \\ 7 & 1 & 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$$

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Vector times vector

Inner product

$\mathbf{a} = (a_1, a_2, \cdots, a_n), \mathbf{b} = (b_1, b_2, \cdots, b_n)$

$$\mathbf{a}\mathbf{b}^{\mathrm{T}} = (a_1, a_2, \cdots, a_n) \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \sum_i a_i b_i$$

$\mathbf{a}\mathbf{b}^{\mathrm{T}} = \mathbf{a} \cdot \mathbf{b} =< \mathbf{a} \cdot \mathbf{b} >$

Quiz: $\mathbf{a} = (2, 3, 1), \mathbf{b} = (2, 4, 6)$

$\mathbf{a} \cdot \mathbf{b} = ?$

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Matrix multiplication

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \mathbf{B} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

$$\mathbf{AB} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

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$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \mathbf{B} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

$$\mathbf{u}_{ab} = (a, b), \mathbf{u}_{cd} = (c, d), \mathbf{v}_{AC} = \begin{pmatrix} A \\ C \end{pmatrix}, \mathbf{v}_{BD} = \begin{pmatrix} B \\ D \end{pmatrix}$$

$$\mathbf{AB} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} -\mathbf{u}_{ab}- \\ -\mathbf{u}_{cd}- \end{pmatrix} \begin{pmatrix} | & | \\ \mathbf{v}_{AC} & \mathbf{v}_{BD} \\ | & | \end{pmatrix}$$

$$= \begin{pmatrix} \mathbf{u}_{ab} \cdot \mathbf{v}_{AC} & \mathbf{u}_{ab} \cdot \mathbf{v}_{BD} \\ \mathbf{u}_{cd} \cdot \mathbf{v}_{AC} & \mathbf{u}_{cd} \cdot \mathbf{v}_{BD} \end{pmatrix}$$

$$= \begin{pmatrix} aA + bC & aB + bD \\ cA + dC & cB + dD \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} Aa + Bc & Ab + Bd \\ Ca + Dc & Cb + Dd \end{pmatrix}$$

$\mathbf{AB} \overset{?}{=} \mathbf{BA}$

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Quiz

$$\begin{pmatrix} 0 & 0 & 2 \\ 0 & -3 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 1 & 4 \\ 2 & 3 \end{pmatrix} = ?$$

$$\begin{pmatrix} 0 & 0 & 2 \\ 0 & -3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 1 & 4 \\ 2 & 3 \end{pmatrix} = ?$$

$$\begin{pmatrix} 1 & 4 \\ 1 & 4 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 0 & 0 & 2 \\ 0 & -3 & 2 \\ 0 & 0 & 0 \end{pmatrix} = ?$$

The rule of multiplication between matrices: $(m \times n)(n \times k) = (m \times k)$

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Matrix and vector multiplication

$$\mathbf{Ax} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Method 1: $\mathbf{a}_1 = \begin{pmatrix} a_{11} & a_{12} \end{pmatrix}$, $\mathbf{a}_2 = \begin{pmatrix} a_{21} & a_{22} \end{pmatrix}$, $\mathbf{A} = \begin{pmatrix} -\mathbf{a}_1- \\ -\mathbf{a}_2- \end{pmatrix}$

$$\mathbf{Ax} = \begin{pmatrix} - & \mathbf{a}_1 & - \\ - & \mathbf{a}_2 & - \end{pmatrix} \begin{pmatrix} | \\ \mathbf{x} \\ | \end{pmatrix}$$

$$= \begin{pmatrix} \begin{pmatrix} a_{11} & a_{12} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ \begin{pmatrix} a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \end{pmatrix} = \begin{pmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{pmatrix}$$

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Matrix and vector multiplication

$$\mathbf{Ax} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Method 2: $\mathbf{a}_1 = \begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix}$, $\mathbf{a}_2 = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$, $\mathbf{A} = \begin{pmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{pmatrix}$

$$\mathbf{Ax} = \begin{pmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$= x\mathbf{a}_1 + y\mathbf{a}_2$$

$$= x \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} + y \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} = \begin{pmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{pmatrix}$$

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